Advanced Computer networks

2025-2026

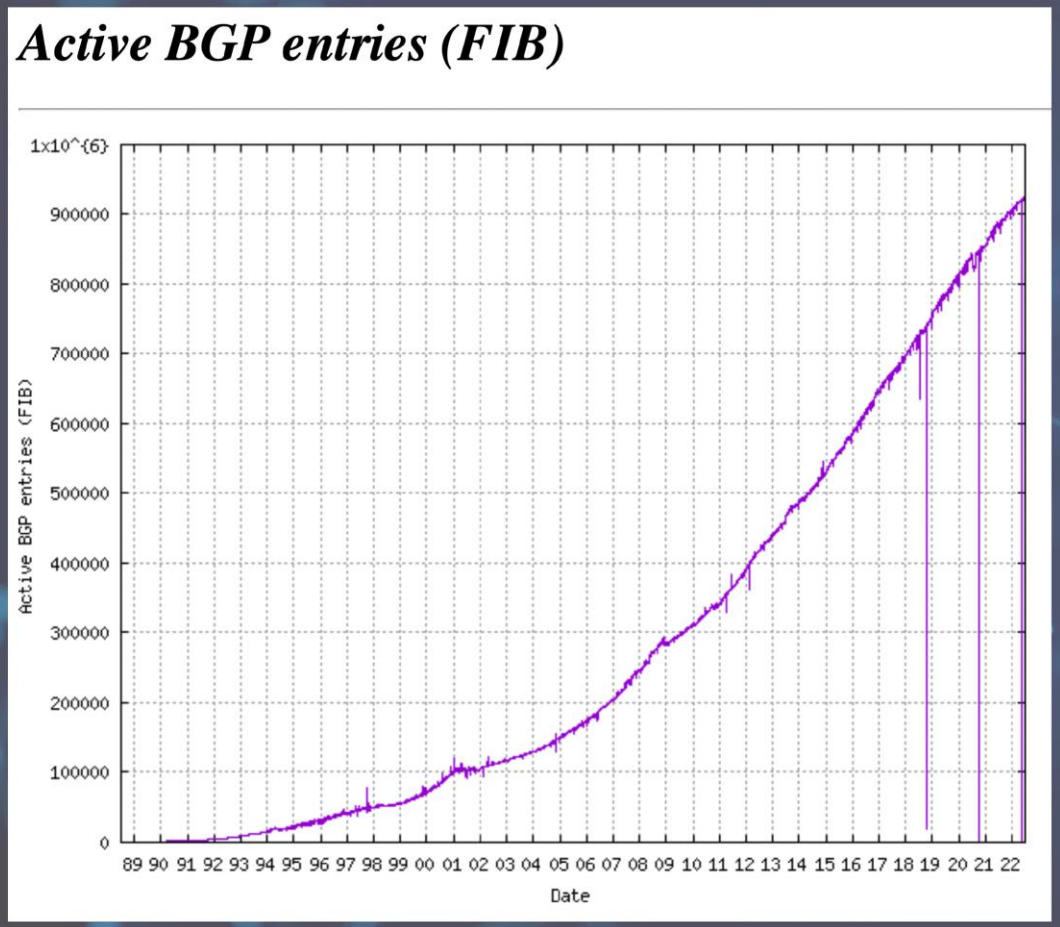
BGP – introduction

Michele Bocca Garetto – based on slides by: kathara.org, Kevin Wallace Training (kwtrain.com)

BGP (Border Gateway Protocol)

- The Exterior Gateway Protocol that holds the Internet together

BGP Path Attributes

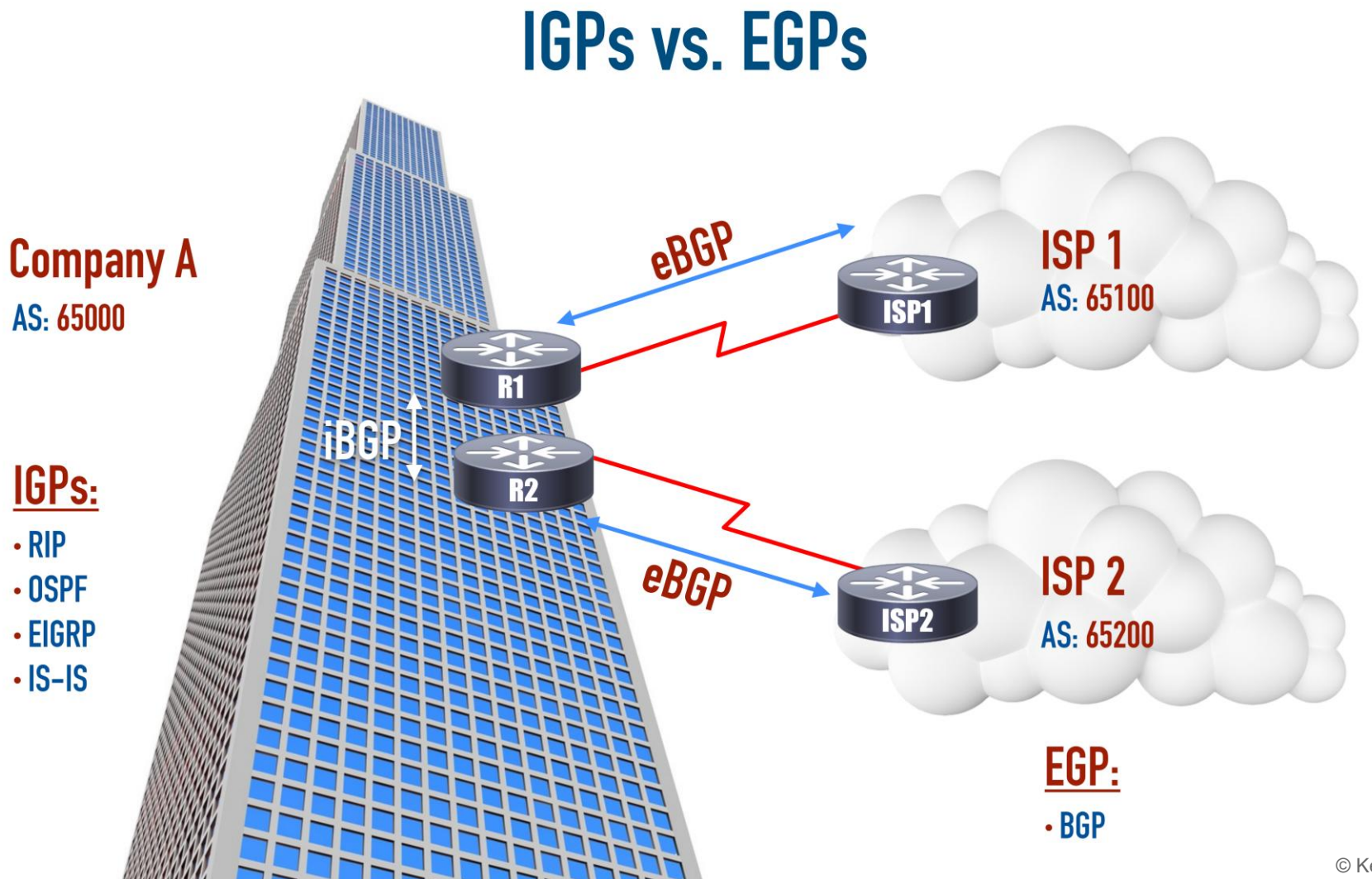


https://www.cidr-report.org

© Kevin Wallace

BGP (Border Gateway Protocol)

- The Exterior Gateway Protocol that holds the Internet together



© Kevin Wallace

iBGP vs eBGP

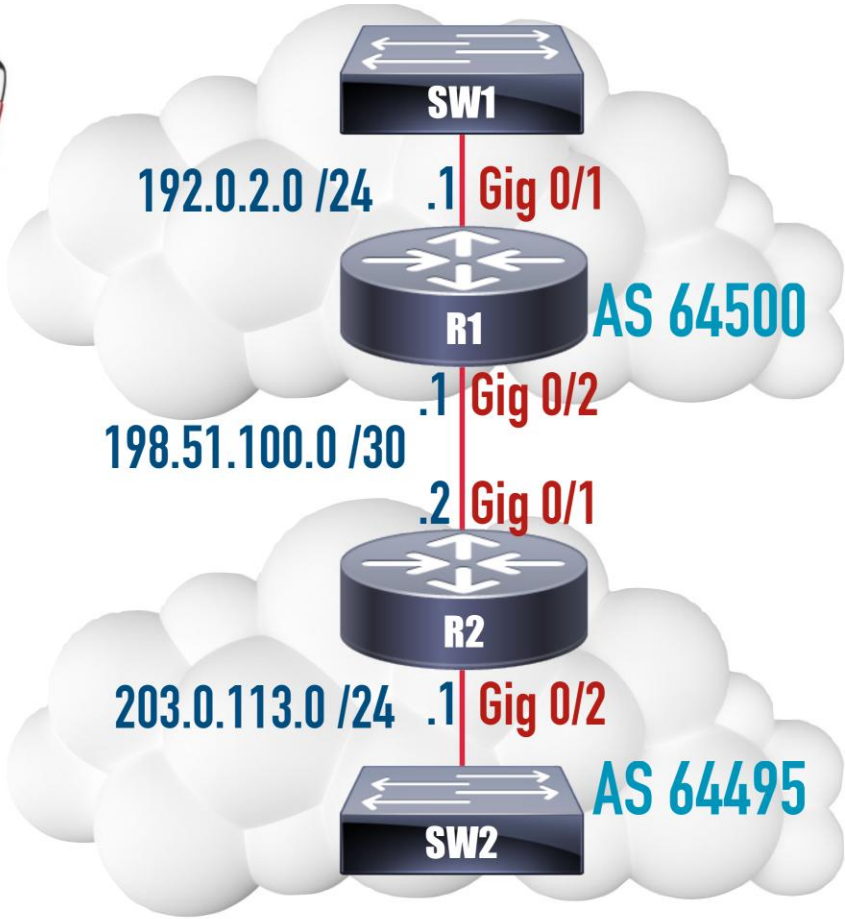
Internal BGP (iBGP): Runs within an autonomous system

External BGP (eBGP): Runs between autonomous systems

Routing Source	Administrative Distance
Connected	0
Static	1 (by default)
eBGP	20
EIGRP (Internal)	90
OSPF	110
IS-IS	115
RIP	120
EIGRP (External)	190
iBGP	200

© Kevin Wallace

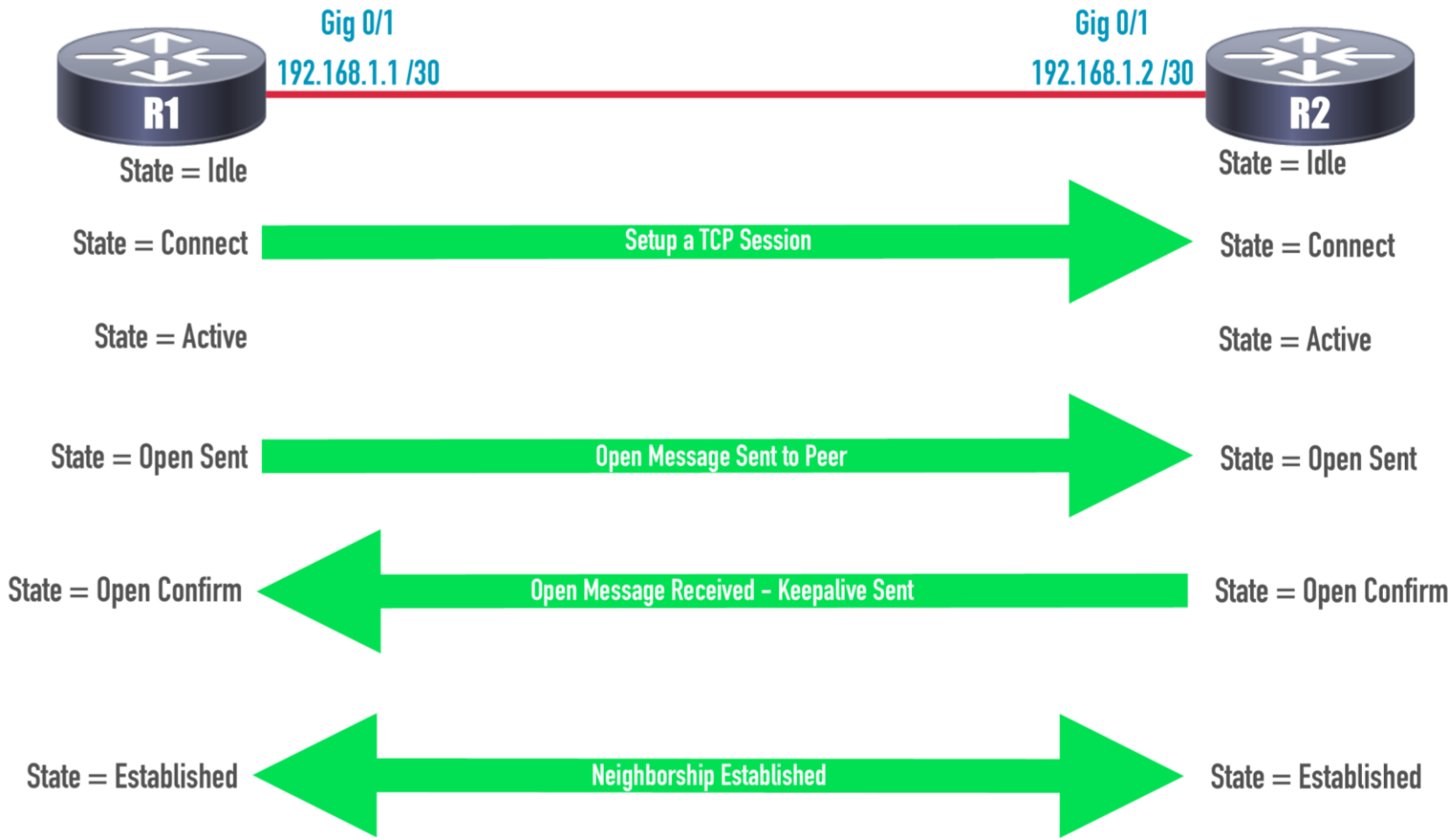
BGP characteristics



© Kevin Wallace

Neighbor formation

BGP Neighbor Formation and States



© Kevin Wallace

BGP Messages and Timers

Open

Includes BGP Version Number, Local AS Number, Hold Time, BGP Router ID, Optional Parameters

Keepalive

A message header that keeps the Holdtime timer from expiring

Update

Can contain withdrawn routes, path attributes, and NLRI

Notification Message

Contains an error code, an error subcode, and information about the error

Default Timers

Keepalive: 60 seconds
Holdtime: 180 Seconds

© Kevin Wallace

Simple peering

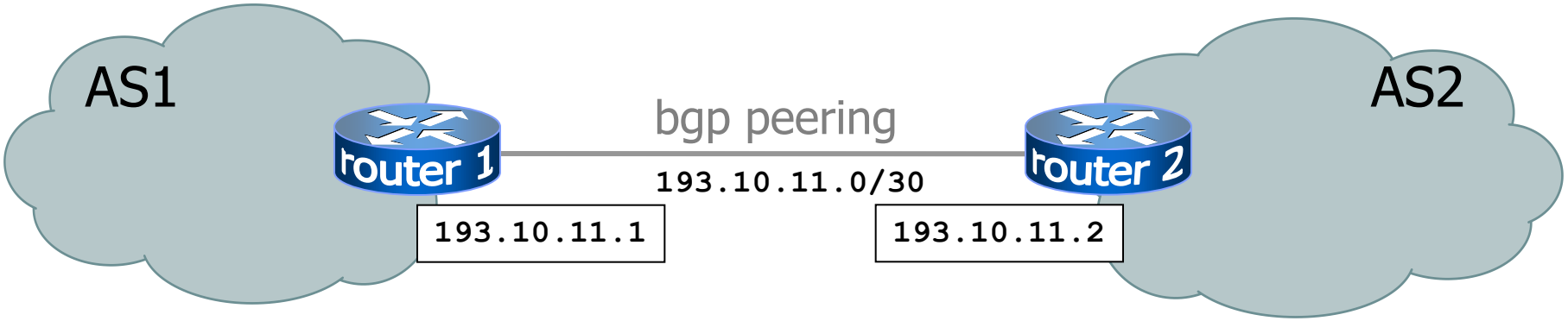
preconditions

- for this lab we assume you have chosen “kathara/frr” as the default image of your Kathará installation
  - execute “kathara settings”
    - select “choose default image”
    - select “kathara/frr”
    - exit from the settings procedure

© Computer Networks
Research Group Roma Tre

A bgp peering between two AS

- a bgp peering is the tcp connection over which routing information will be exchanged



© Computer Networks
Research Group Roma Tre

peering configuration commands

command syntax

```
! <a-comment-on-a-single-line>
```

command syntax

```
router bgp <my-as-number>
```

command syntax

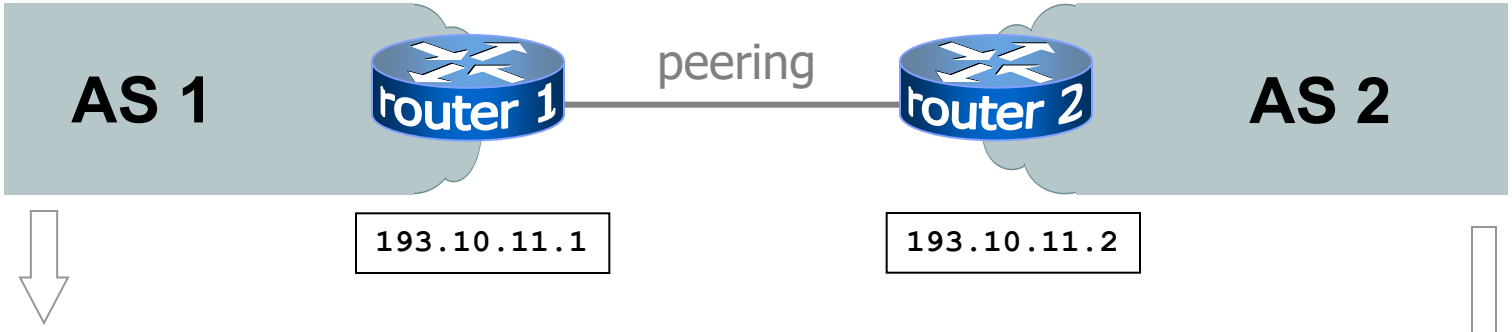
```
neighbor <neighbor-ip> remote-as <neighbor-as-num>
```

command syntax

```
neighbor <neighbor-ip> description <text>
```

© Computer Networks
Research Group Roma Tre

peering configuration example

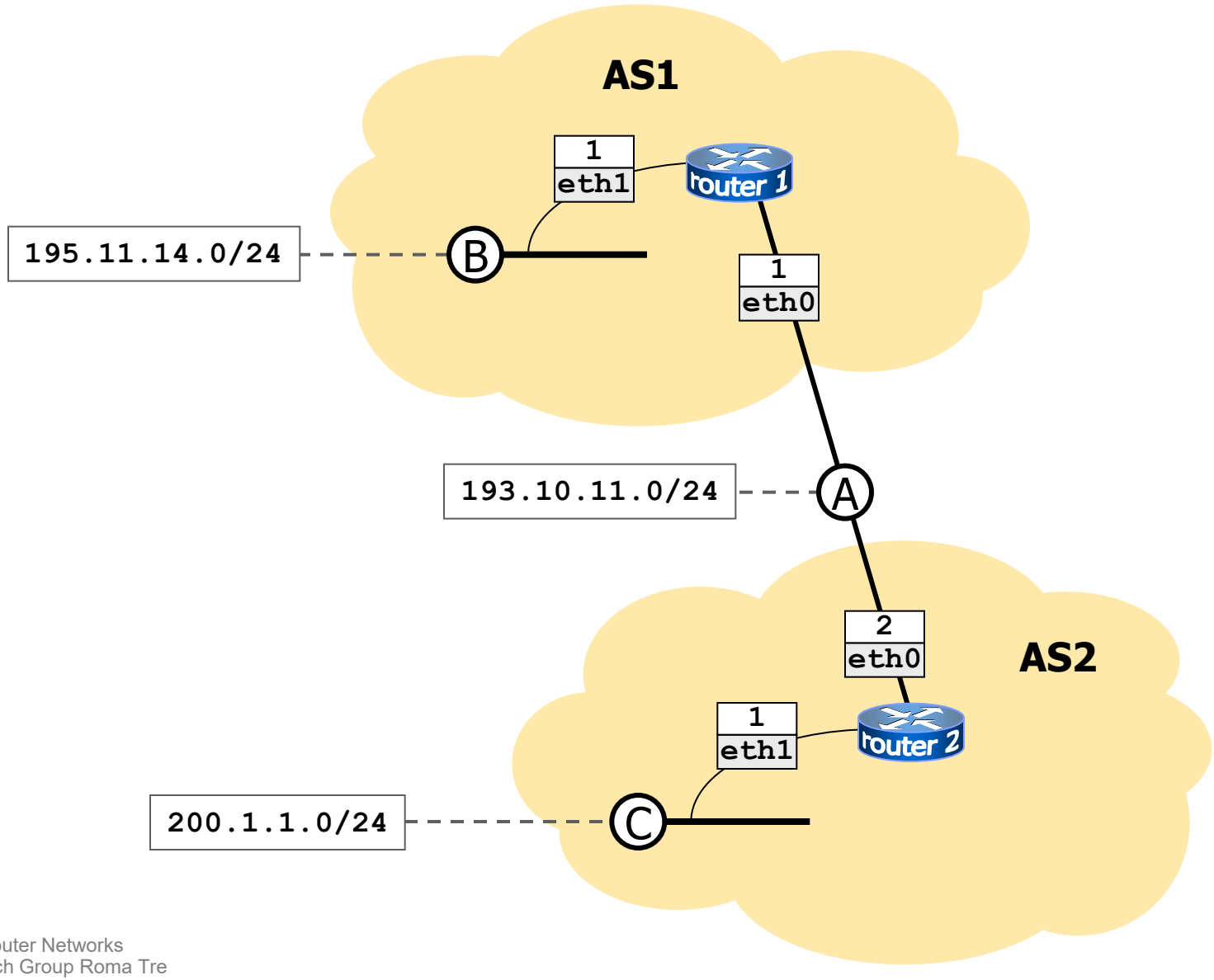



```
! router 1 configuration file
router bgp 1
neighbor 193.10.11.2 remote-as 2
neighbor 193.10.11.2 description Router 2
```

```
! router 2 configuration file
router bgp 2
neighbor 193.10.11.1 remote-as 1
neighbor 193.10.11.1 description Router 1
```

© Computer Networks
Research Group Roma Tre

a simple peering



© Computer Networks
Research Group Roma Tre

a simple peering

- launch the script

host machine

```
user@localhost:~$ cd kathara-lab_bgp-simple-peering_frr
user@localhost:~/kathara-lab_bgp-simple-peering_frr$ kathara lstart
```

- check the frr configuration file

router1

```
router1:~# less /etc/frr/frr.conf
!
password zebra
enable password zebra
!
router bgp 1
...
...
```

© Computer Networks
Research Group Roma Tre

a simple peering

- check the kernel routing table

router1

```
router1:~# route
Kernel IP routing table
Destination     Gateway         Genmask         Flags Metric Ref    Use Iface
193.10.11.0     *               255.255.255.0   U     0      0        0 eth0
195.11.14.0     *               255.255.255.0   U     0      0        0 eth1
router1:~#
```

  - as no routing protocol (not even bgp!) is propagating routing information, only local routes are known

© Computer Networks
Research Group Roma Tre

a simple peering

- check the log file of the frr daemons

router1

```
root@router1:/# cat /var/log/frr/frr.log
2021/10/23 21:31:28 ZEBRA: client 30 says hello and bids fair to announce only vnc routes vrf=0
2021/10/23 21:31:28 ZEBRA: client 26 says hello and bids fair to announce only bgp routes vrf=0
2021/10/23 21:31:28 ZEBRA: client 37 says hello and bids fair to announce only static routes vrf=0
2021/10/23 21:31:29 BGP: 193.10.11.2 sending KEEPALIVE
2021/10/23 21:31:29 BGP: 193.10.11.2 KEEPALIVE rcvd
2021/10/23 21:31:30 BGP: u1:s1 announcing routes upon coalesce timer expiry(1050 ms)
2021/10/23 21:31:30 BGP: 193.10.11.2 rcvd UPDATE wlen 0 attrlen 0 alen 0
2021/10/23 21:31:30 BGP: bgp_best_path_select_defer: processing route for IPv4 Unicast : cnt 0
2021/10/23 21:31:30 BGP: bgp_update_receive: rcvd End-of-RIB for IPv4 Unicast from 193.10.11.2 in vrf default
2021/10/23 21:32:29 BGP: 193.10.11.2 sending KEEPALIVE
2021/10/23 21:32:29 BGP: 193.10.11.2 KEEPALIVE rcvd
2021/10/23 21:33:29 BGP: 193.10.11.2 sending KEEPALIVE
2021/10/23 21:33:29 BGP: 193.10.11.2 KEEPALIVE rcvd
root@router1:/#
```

© Computer Networks
Research Group Roma Tre

a simple peering

- check the command line interface of bgpd

router1

```
root@router1:/# telnet localhost bgpd
Trying 127.0.0.1...
Connected to localhost.
Escape character is '^]'.

Hello, this is FRRouting (version 7.5.1).
Copyright 1996-2005 Kunihiro Ishiguro, et al.

User Access Verification

Password: zebra
router1> show ip bgp summary

IPv4 Unicast Summary:
BGP router identifier 195.11.14.1, local AS number 1 vrf-id 0
BGP table version 0
RIB entries 0, using 0 bytes of memory
Peers 1, using 21 KiB of memory

Neighbor        V         AS   MsgRcvd   MsgSent   TblVer  InQ OutQ  Up/Down State/PfxRcd   PfxSnt
193.10.11.2     4          2         9         9        0    0    0 00:06:06            0        0

Total number of neighbors 1
router1>
```

© Computer Networks
Research Group Roma Tre

a simple peering

- check the peering status

router1

```
router1> show ip bgp neighbors
BGP neighbor is 193.10.11.2, remote AS 2, local AS 1, external link
 Description: Router 2 of AS2
Hostname: router2
  BGP version 4, remote router ID 200.1.1.1, local router ID 195.11.14.1
  BGP state = Established, up for 00:08:31
  Last read 00:00:31, Last write 00:00:31
  Hold time is 180, keepalive interval is 60 seconds
  Neighbor capabilities:
    4 Byte AS: advertised and received
    AddPath:
      IPv4 Unicast: RX advertised IPv4 Unicast and received
    Route refresh: advertised and received(old & new)
    Address Family IPv4 Unicast: advertised and received
    Hostname Capability: advertised (name: router1,domain name: n/a) received
(name: router2,domain name: n/a)
    Graceful Restart Capability: advertised and received
      Remote Restart timer is 120 seconds
      Address families by peer:
        none
  Graceful restart information:
    End-of-RIB send: IPv4 Unicast
    End-of-RIB received: IPv4 Unicast
    Local GR Mode: Helper*
 --More--
```

© Computer Networks
Research Group Roma Tre

a simple peering

- show the bgp table

router1

```
router1-frr# show ip bgp
No BGP prefixes displayed, 0 exist
router1-frr#
```

- stop the lab

host machine

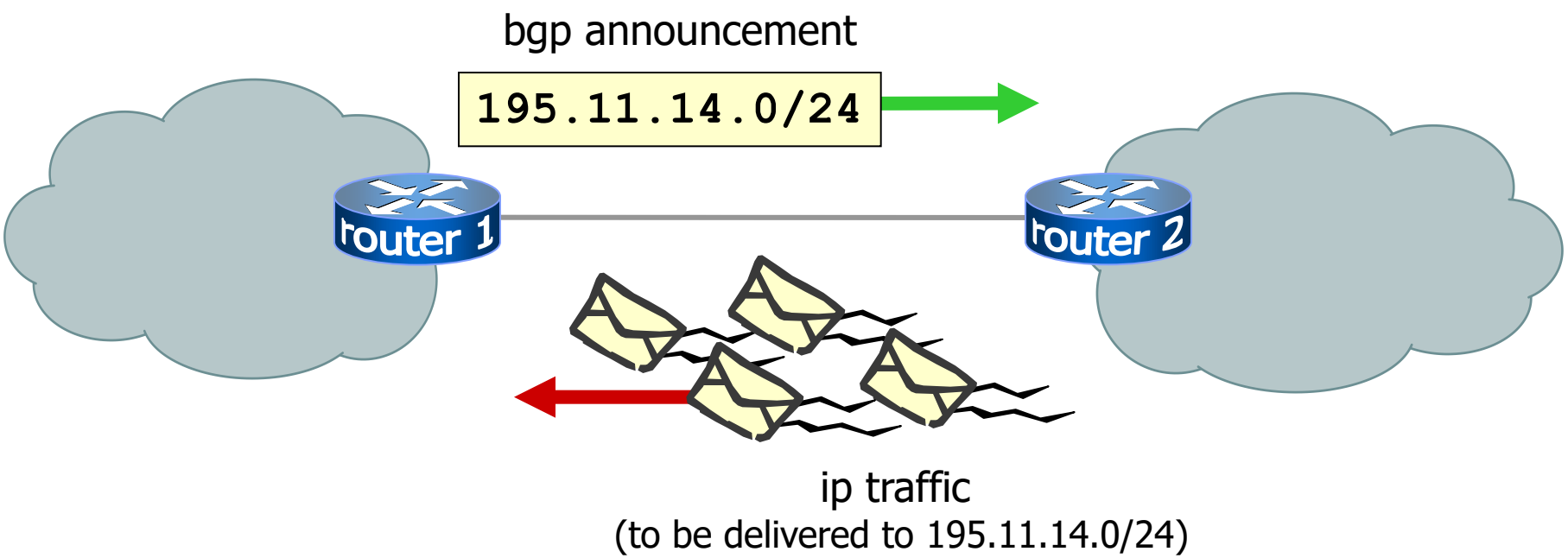
```
user@localhost:~$ cd kathara-lab_bgp-simple-peering_frr
user@localhost:~/kathara-lab_bgp-simple-peering_frr$ kathara lclean
```

© Computer Networks
Research Group Roma Tre

Bgp announcements

announcements and traffic flows

- bgp allows a router to offer connectivity to another router
- "offering connectivity" means "promising the delivery to a specific destination"



© Computer Networks
Research Group Roma Tre

announcement commands

cisco command syntax

```
network <network-ip> mask <network-mask>
```

frr command syntax

```
network <network-ip/network-mask>
```

- the network command performs the following
  - checks whether the current router has the specified prefix in its kernel forwarding table
    - if not, the command has no effect whatsoever
  - for each peer checks whether the BGP configuration has security filters associated with the peer
    - only in the positive case the prefix is announced to the peer

© Computer Networks
Research Group Roma Tre

announcement commands

command syntax

```
no bgp network import-check
```

- this command modifies the network statement disabling the check of the existence of the prefix in the kernel forwarding table
  - default value is **`bgp network import-check`**
- by disabling this check you may announce prefixes that
  - may not be local
  - may not even exist (!)
- not needed for this lab

© Computer Networks
Research Group Roma Tre

announcement commands

command syntax

```
no bgp ebgp-requires-policy
```

- the used FRR BGP version is RFC-8212 compliant
  - requires incoming and outgoing filters to be applied for eBGP sessions
  - without incoming filters, no routes will be accepted
  - without outgoing filters, no routes will be announced
- the command modifies the network statement disabling the check of the existence of incoming and outgoing filters

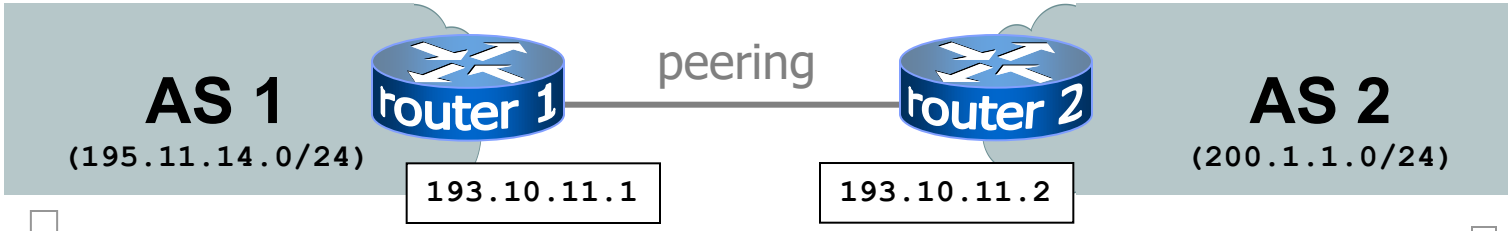
© Computer Networks
Research Group Roma Tre

announcement commands

- observe that the **`network`** command
  - *does not* inject any route in the kernel forwarding table
  - checks whether the network address matches the netmask; if it does not, the command is automatically replaced in the router configuration; for example:
    - **`network 193.100.0.0/8`**
      is replaced by
      `network 193.0.0.0/8`
    - **`network 1.2.3.4/0`**
      is replaced by
      `network 0.0.0.0/0`

© Computer Networks
Research Group Roma Tre

announcement example

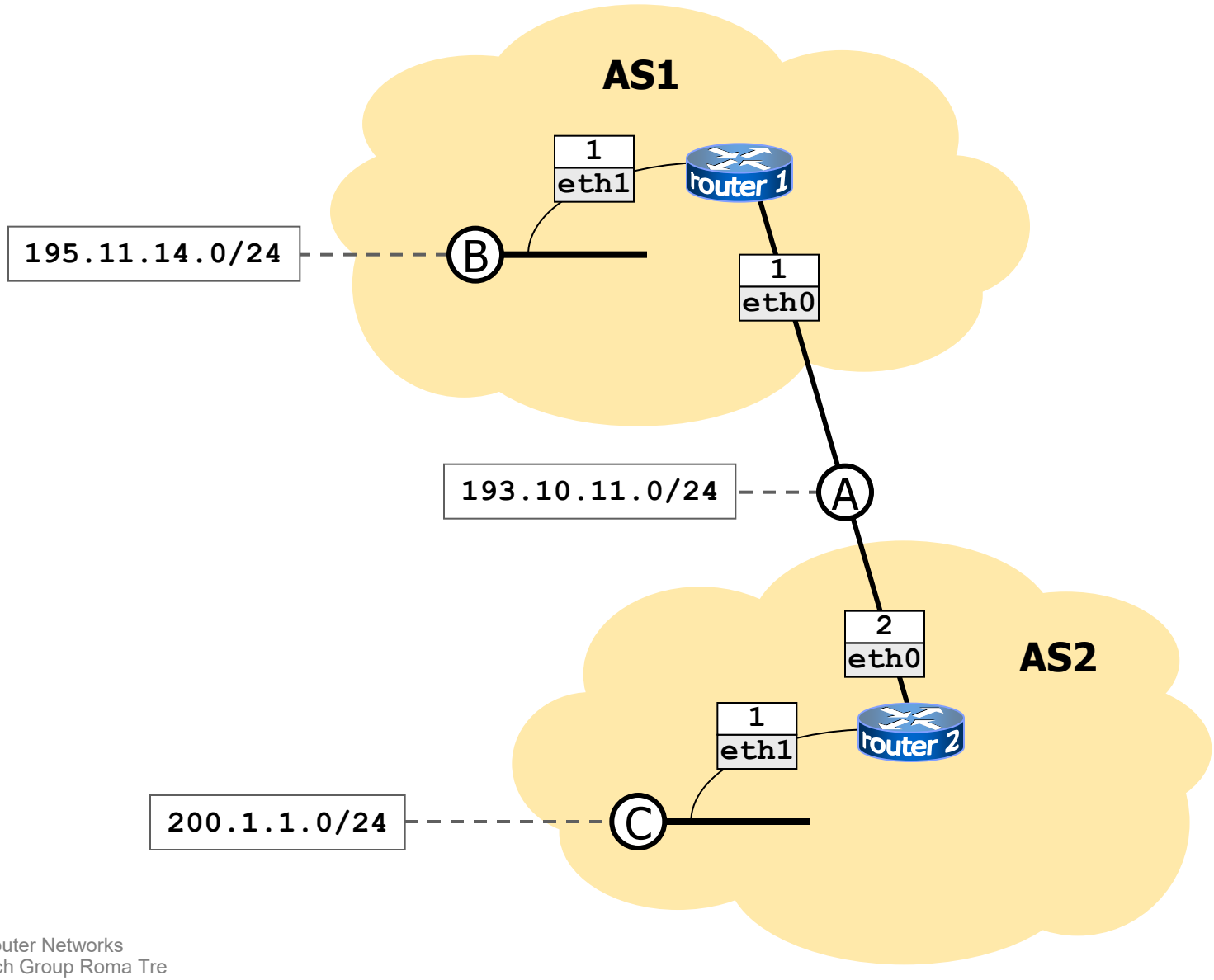



```
! router 1 configuration file
router bgp 1
no bgp ebgp-requires-policy
neighbor 193.10.11.2 remote-as 2
network 195.11.14.0/24
```

```
! router 2 configuration file
router bgp 2
no bgp ebgp-requires-policy
neighbor 193.10.11.1 remote-as 1
network 200.1.1.0/24
```

© Computer Networks
Research Group Roma Tre

peering configuration



© Computer Networks
Research Group Roma Tre

announcement example

- start the lab

host machine

```
user@localhost:~$ cd kathara-lab_bgp-announcement_frr
user@localhost:~/kathara-lab_bgp-announcement_frr$ kathara lstart
```

- check the zebra routing table

router2

```
root@router2:/# vtysh
...
router2-frr# show ip route
Codes: K - kernel route, C - connected, S - static, R - RIP,
       O - OSPF, I - IS-IS, B - BGP, E - EIGRP, N - NHRP,
       T - Table, v - VNC, V - VNC-Direct, A - Babel, D - SHARP,
       F - PBR, f - OpenFabric,
       > - selected route, * - FIB route, q - queued, r - rejected, b -
backup

C>* 193.10.11.0/24 is directly connected, eth0, 00:03:41
B>* 195.11.14.0/24 [20/0] via 193.10.11.1, eth0, weight 1, 00:03:38
C>* 200.1.1.0/24 is directly connected, eth1, 00:03:41
router2-frr#
```

© Computer Networks
Research Group Roma Tre

announcement example

- check the bgpd log file

router2

```
root@router2:/# cat /var/log/frr/frr.log
...
2021/10/23 22:44:06 BGP: 193.10.11.1 sending KEEPALIVE
2021/10/23 22:44:06 BGP: 193.10.11.1 KEEPALIVE rcvd
2021/10/23 22:44:07 BGP: u1:s1 announcing routes upon coalesce timer expiry(1050 ms)
2021/10/23 22:44:07 BGP: u1:s1 send UPDATE w/ attr: nexthop 0.0.0.0, origin i,
mp_nexthop ::, metric 0, path
2021/10/23 22:44:07 BGP: u1:s1 send UPDATE 200.1.1.0/24 IPv4 unicast
2021/10/23 22:44:07 BGP: u1:s1 send UPDATE len 55 numpfx 1
2021/10/23 22:44:07 BGP: u1:s1 193.10.11.1 send UPDATE w/ nexthop 193.10.11.2
2021/10/23 22:44:07 BGP: 193.10.11.1 rcvd UPDATE w/ attr: nexthop 193.10.11.1, origin
i, metric 0, path 1
2021/10/23 22:44:07 BGP: 193.10.11.1 rcvd UPDATE wlen 0 attrlen 28 alen 4
2021/10/23 22:44:07 BGP: 193.10.11.1 rcvd 195.11.14.0/24 IPv4 unicast
2021/10/23 22:44:07 BGP: 193.10.11.1 rcvd UPDATE wlen 0 attrlen 0 alen 0
2021/10/23 22:44:07 BGP: bgp_best_path_select_defer: processing route for IPv4 Unicast
: cnt 0
2021/10/23 22:44:07 BGP: bgp_update_receive: rcvd End-of-RIB for IPv4 Unicast from
193.10.11.1 in vrf default
...
```

© Computer Networks
Research Group Roma Tre

announcement example

- check the bgpd log file

router2

```
root@router2:/# cat /var/log/frr/frr.log
...
2021/10/23 22:44:06 BGP: 193.10.11.1 sending KEEPALIVE
2021/10/23 22:44:06 BGP: 193.10.11.1 KEEPALIVE rcvd
2021/10/23 22:44:07 BGP: u1:s1 announcing routes upon co[illegible]ce timer expiry(1050 ms)
2021/10/23 22:44:07 BGP: u1:s1 send UPDATE w/ attr: ne[illegible]op 0.0.0.0, origin i,
mp_nexthop ::, metric 0, path
202[illegible]s1 send UPDATE 200.1.1.0/24 IPv4 unicast
202[illegible]1 send UPDATE len 55 numpfx 1
202[illegible]1 193.10.11.1 send UPDATE w/ nexthop 193.10.11.2
202[illegible]0.11.1 rcvd UPDATE w/ attr: nexthop 193.10.11.1, origin
i,
2021/[illegible] 193.10.11.1 rcvd UPDATE wlen 0 attrlen 28 alen 4
2021/10/23 22:44:07 BGP: 193.10.11.1 rcvd 195.11.14.0/24 IPv4 unicast
2021/10/23 22:44:07 BGP: 193.10.11.1 rcvd UPDATE wlen 0 attrlen 0 alen 0
2021/10/23 22:44:07 BGP: bgp_best_path_select_defer: processing route for IPv4 Unicast
: cnt 0
2021/10/23 22:44:07 BGP: bgp_update_receive: rcvd End-of-RIB for IPv4 Unicast from
193.10.11.1 in vrf default
...
```

sent announcement

received announcement

© Computer Networks
Research Group Roma Tre

announcement example

- check the vtysh cli (command line interface)
  - type "**`vtysh`**"
  - type "**`show ip bgp neighbors`**"
  - type "**`show ip bgp`**"
  - type "**`show ip bgp 200.1.1.0`**"
- ping "**`200.1.1.0`**"
- terminate the lab

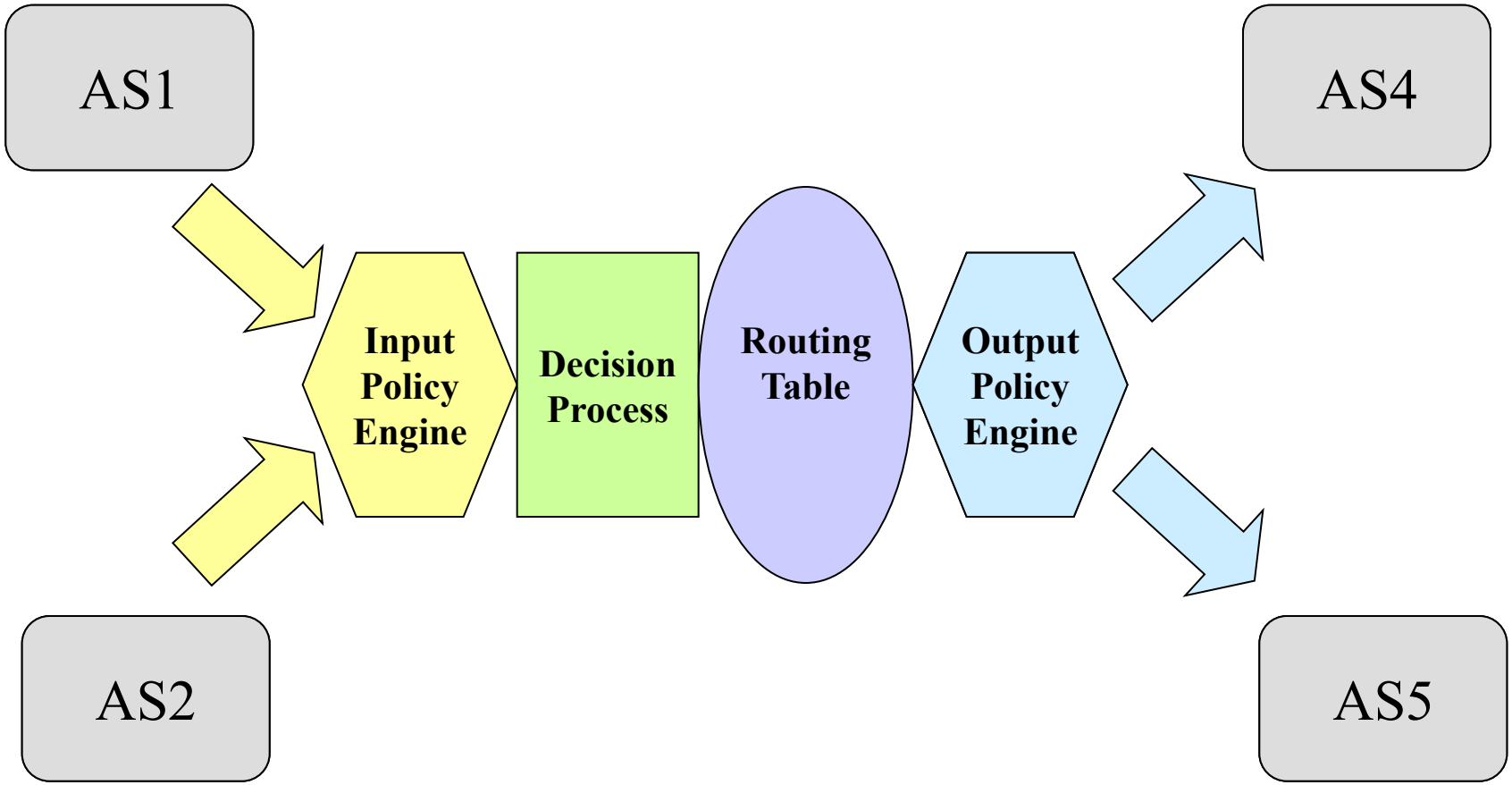
host machine

```
user@localhost:~$ cd kathara-lab_bgp-announcement_frr
user@localhost:~/kathara-lab_bgp-announcement_frr$ kathara lclean
```

© Computer Networks
Research Group Roma Tre

BGP filtering policies

BGP decision process and architecture



© Computer Networks
Research Group Roma Tre

applying policies

1. announcement filtering
   - send/accept an announcement only if some condition is verified
   - commands:
     - **`prefix-list`** used to filter prefixes
     - **`filter-list`** used to filter as numbers
2. announcement tuning
   - attach to your announcement some information (attributes) that should be considered by the receiver
   - commands:
     - **`route-map`**
     - **`access-list`** used to match prefixes or as-paths in a **`route-map`**

© Computer Networks
Research Group Roma Tre

prefix filtering (prefix-list)

© Computer Networks
Research Group Roma Tre

prefix filtering commands

command syntax

```
neighbor <neighbor-ip> prefix-list <p-list-name> in
```

command syntax

```
neighbor <neighbor-ip> prefix-list <p-list-name> out
```

command syntax

```
ip prefix-list <p-list-name> permit <network/mask>
```

command syntax

```
ip prefix-list <p-list-name> deny <network/mask>
```

© Computer Networks
Research Group Roma Tre

prefix filtering: example

frr configuration file

```
router bgp 1
! no bgp ebgp-requires-policy (not needed anymore)
neighbor 193.10.11.2 remote-as 2
neighbor 193.10.11.2 description Router 2 of AS2
!
network 195.11.14.0/24
network 195.11.15.0/24
neighbor 193.10.11.2 prefix-list partialIn in
neighbor 193.10.11.2 prefix-list partialOut out
!
ip prefix-list partialOut seq 5 permit 195.11.14.0/24
ip prefix-list partialIn seq 5 deny 200.1.1.0/24
ip prefix-list partialIn seq 10 permit any
```

only `195.11.14.0/24` is announced to neighbor `193.10.11.2`
all with the exception of `200.1.1.0/24` is accepted from `193.10.11.2`

© Computer Networks
Research Group Roma Tre

about `prefix-lists`

- **`prefix-list`** entries are ordered according to a sequence number
  - explicitly assigned by the user; example:
    - `ip prefix-list myPfxList seq 5 permit 10.0.0.0/8`
  - implicitly assigned by frr; example:
    - `ip prefix-list myPfxList permit 10.0.0.0/8`
      `ip prefix-list myPfxList permit 20.0.0.0/8`
      is automatically turned to:
      `ip prefix-list myPfxList seq 5 permit 10.0.0.0/8`
      `ip prefix-list myPfxList seq 10 permit 20.0.0.0/8`

© Computer Networks
Research Group Roma Tre

about `prefix-lists`

- the first matching entry is applied; example:
  - `ip prefix-list letThru permit 10.0.0.0/8`
    `ip prefix-list letThru deny any`
    accepts `10.0.0.0/8` only
  - `ip prefix-list throwAway deny any`
    `ip prefix-list throwAway permit 10.0.0.0/8`
    rejects everything

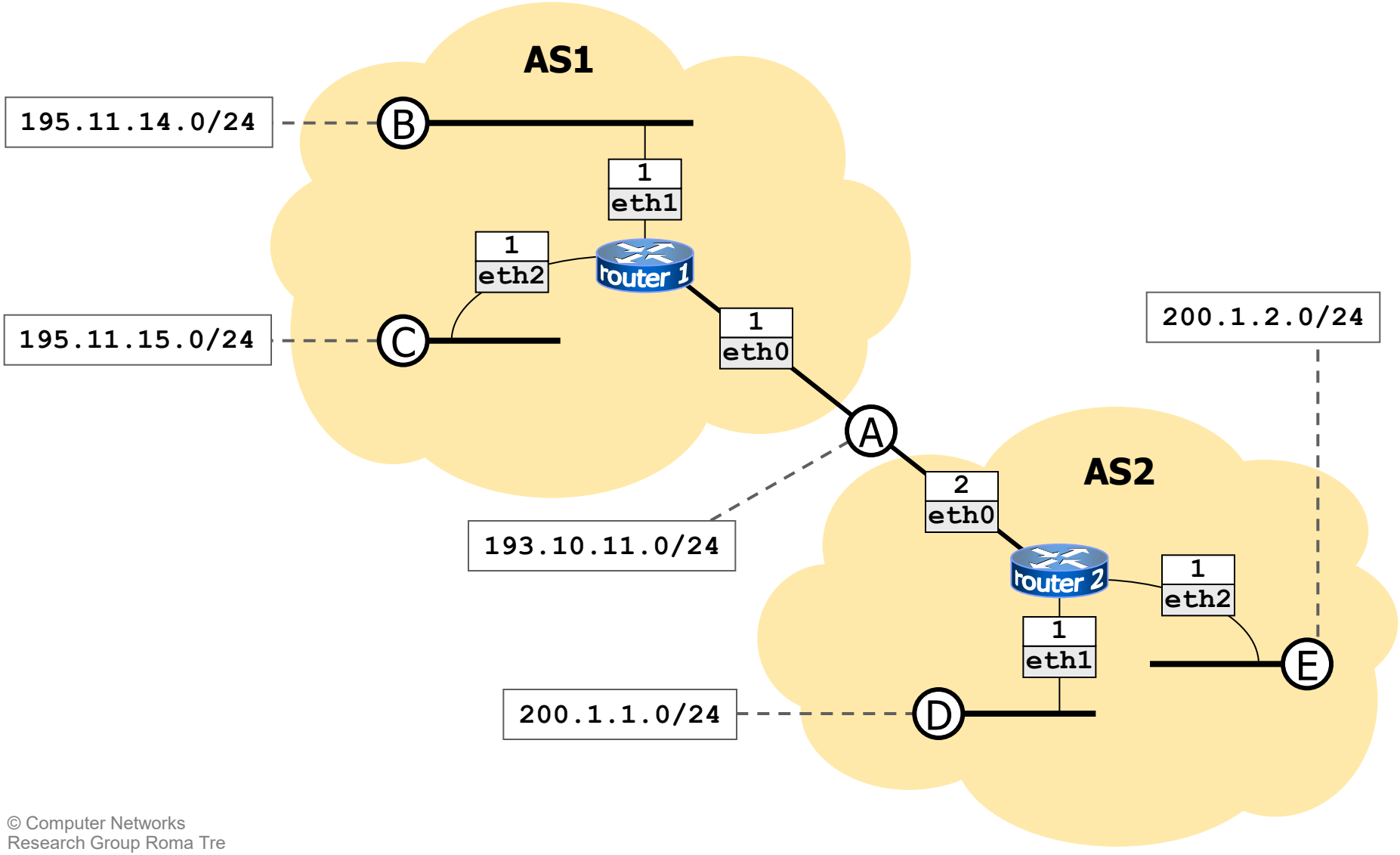
© Computer Networks
Research Group Roma Tre

`prefix-list` defaults

- in frr, **`prefix-lists`** default to **`deny`**; for example:
  - **`ip prefix-list myPrefixList permit 10.0.0.0/8`** filters out everything but **`10.0.0.0/8`**
  - **`ip prefix-list myPrefixList deny 10.0.0.0/8`** filters out everything
- referencing an undefined **`prefix-list`** in a **`neighbor`** statement is equivalent to **`deny`**ing **`any`**thing; for example:
  - **`neighbor 10.0.0.1 prefix-list undefinedPrefixList in`** filters out everything if **`undefinedPrefixList`** is not defined

© Computer Networks
Research Group Roma Tre

prefix filtering



© Computer Networks
Research Group Roma Tre

prefix filtering

- start the lab

host machine

```
user@localhost:~$ cd kathara-lab_bgp-prefix-filtering_frr
user@localhost:~/kathara-lab_bgp-prefix-filtering_frr$ kathara lstart
```

- check the frr configuration file

router1

```
router1:~# less /etc/frr/frr.conf
```

- check the frr log file

router1

```
router1:~# less /var/log/frr/frr.log
```

© Computer Networks
Research Group Roma Tre

© Computer Networks
Research Group Roma Tre

prefix filtering

- check the routing table

router1

```
root@router1:/# route
Kernel IP routing table
Destination     Gateway         Genmask         Flags Metric Ref    Use Iface
193.10.11.0     0.0.0.0         255.255.255.0   U     0      0        0 eth0
195.11.14.0     0.0.0.0         255.255.255.0   U     0      0        0 eth1
195.11.15.0     0.0.0.0         255.255.255.0   U     0      0        0 eth2
200.1.2.0       193.10.11.2     255.255.255.0   UG    20     0        0 eth0
root@router1:/# vtysh

Hello, this is FRRouting (version 7.5.1).
Copyright 1996-2005 Kunihiro Ishiguro, et al.

router1-frr# show ip route
Codes: K - kernel route, C - connected, S - static, R - RIP,
       O - OSPF, I - IS-IS, B - BGP, E - EIGRP, N - NHRP,
       T - Table, v - VNC, V - VNC-Direct, A - Babel, D - SHARP,
       F - PBR, f - OpenFabric,
       > - selected route, * - FIB route, q - queued, r - rejected, b - backup

C>* 193.10.11.0/24 is directly connected, eth0, 00:08:56
C>* 195.11.14.0/24 is directly connected, eth1, 00:08:56
C>* 195.11.15.0/24 is directly connected, eth2, 00:08:56
B>* 200.1.2.0/24 [20/0] via 193.10.11.2, eth0, weight 1, 00:08:54
router1-frr#
```

prefix filtering

- check the frr cli (command line interface)

router1

```
router1-frr# show ip bgp neighbors
BGP neighbor is 193.10.11.2, remote AS 2, local AS 1, external link
 Description: Router 2 of AS2
Hostname: router2
  BGP version 4, remote router ID 200.1.2.1,
                 local router ID 195.11.15.1
  BGP state = Established, up for 00:11:36
  Last read 00:00:36, Last write 00:00:36
  Hold time is 180, keepalive interval is 60 seconds
  Neighbor capabilities:
    4 Byte AS: advertised and received
    AddPath:
      IPv4 Unicast: RX advertised IPv4 Unicast and received
    Route refresh: advertised and received(old & new)
    Address Family IPv4 Unicast: advertised and received
...
...
router1-frr# show ip bgp 200.1.1.0
% Network not in table
router1-frr#
```

© Computer Networks
Research Group Roma Tre

bgp attributes

© Computer Networks
Research Group Roma Tre

attributes

- a bgp announcement is a “bag” of attributes
- attributes may be
  - “well-known” or optional
    - well-known attributes are understood by any bgp4 speaker
  - mandatory or discretionary
    - mandatory attributes must be present in updates
  - transitive or nontransitive
    - transitive attributes are passed when received
    - nontransitive attributes traverse a single peering

© Computer Networks
Research Group Roma Tre

attribute list

- prefix
  - the section of ip space announced
- as-path
  - the sequence of traversed ases
- origin
  - igp (route is interior to the originating as)
  - egp (route learned via the egp protocol)
  - incomplete (route learned in some other way)
- next-hop
  - to be inserted in the routing table
- metric (multi-exit-discriminator)
  - asking another as to prefer lower values of it
- local-pref
  - prefer higher values
- atomic aggregate
- aggregator
- weight
  - cisco proprietary

© Computer Networks
Research Group Roma Tre

well-known attributes

- mandatory well-known
  - as-path: the sequence of traversed ASes
  - next-hop: to be inserted in the routing table; in i-bgp stays unchanged
  - origin
- discretionary well-known
  - local preference: asking i-bgp peers to prefer higher values of it
  - atomic aggregate

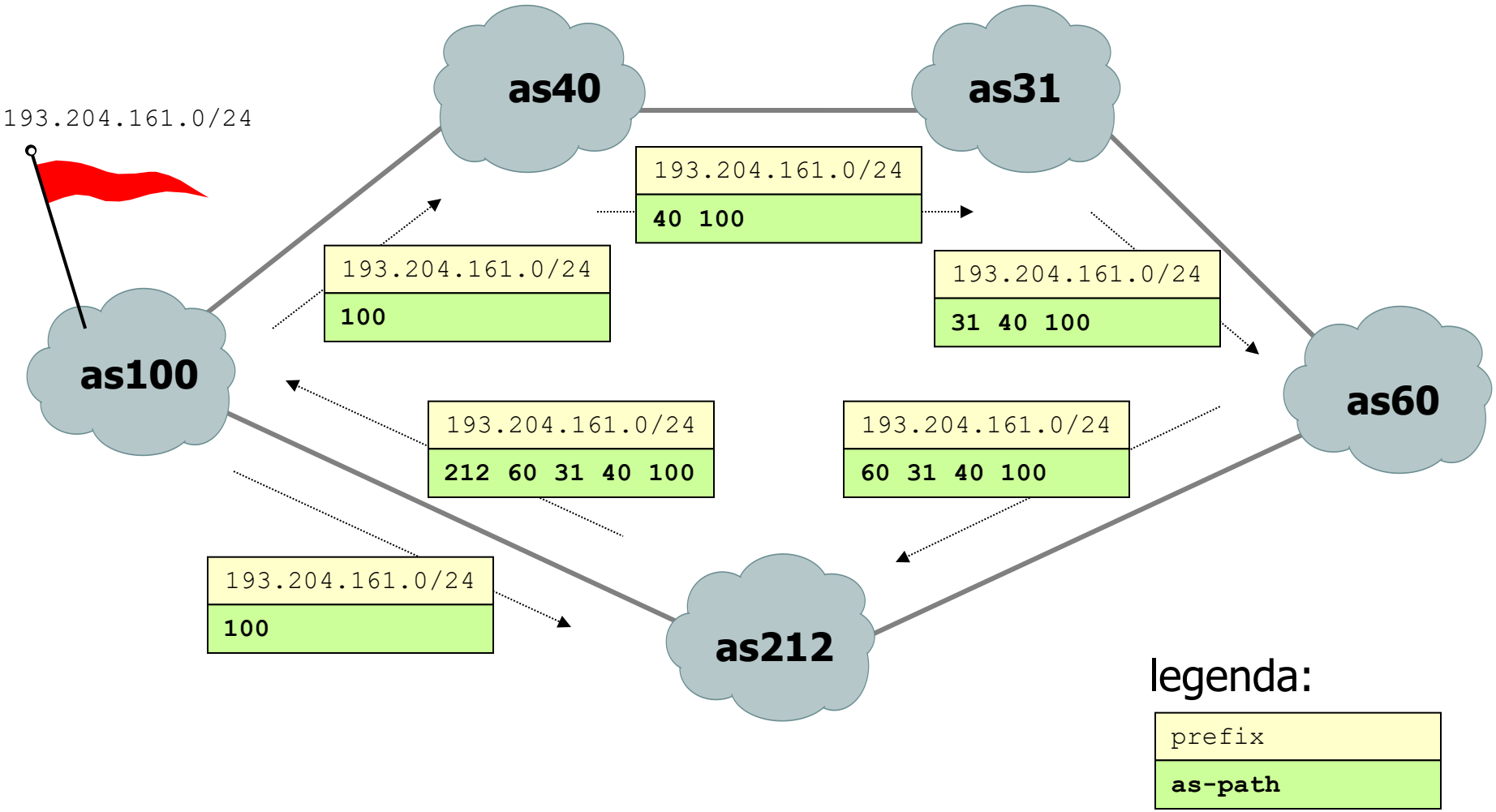
© Computer Networks
Research Group Roma Tre

optional attributes

- non transitive optional
  - multi-exit discriminator: asking other ASes to prefer lower values of it
- transitive optional
  - aggregator
  - community

© Computer Networks
Research Group Roma Tre

attributes: prefix & as-path



© Computer Networks
Research Group Roma Tre

attributes: as-path

- the as-path is empty when a local route is inserted into the BGP table
- when the announcement goes to a different as (e-BGP) the as number is inserted at the end of the as-path (prepending)
  - a router knows which and how many ases should be traversed to reach the destination
  - loops are avoided
  - policies can be applied
- in i-BGP the as-path does not change

© Computer Networks
Research Group Roma Tre

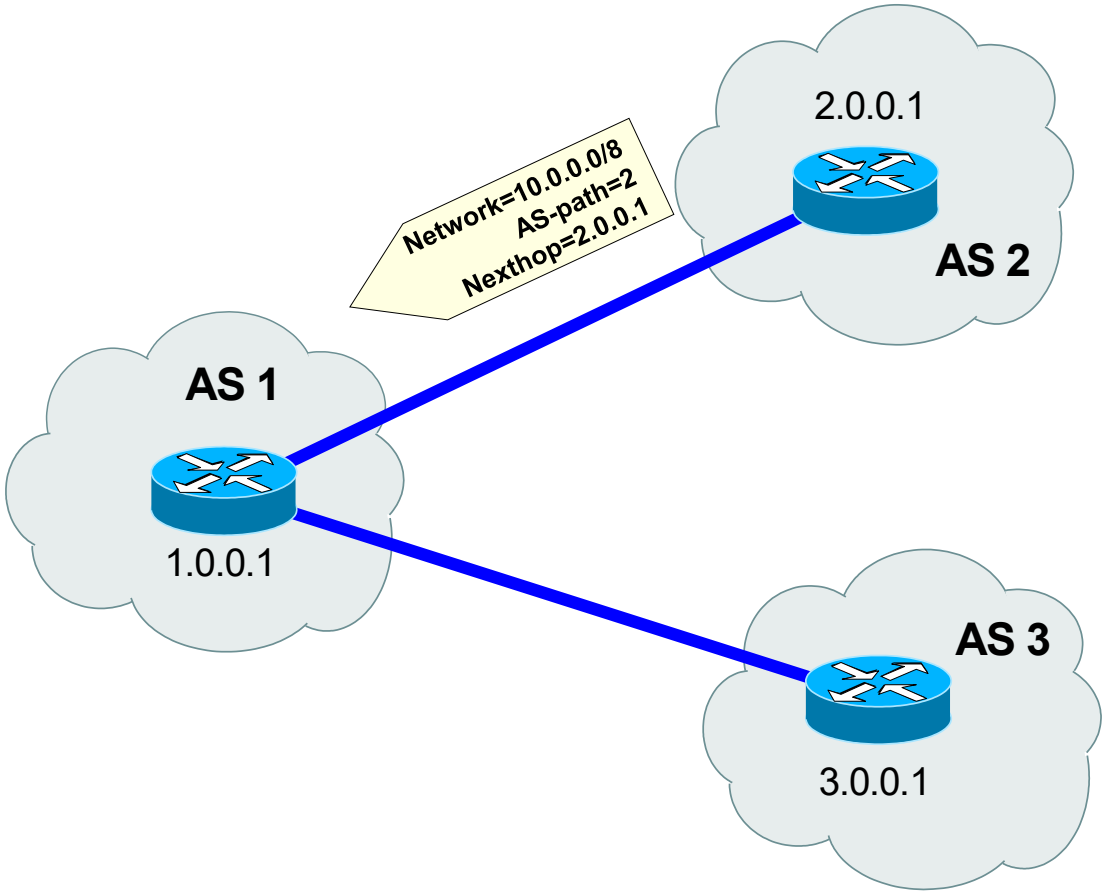
attributes: nexthop

- where to send packets for a specific ip network
- usually, the nexthop is the router that sends the announcements
  - exceptions:
    - "*shared media*" (ethernet, etc..)
    - i-BGP announcements of networks learned using e-BGP

© Computer Networks
Research Group Roma Tre

attributes: nexthop

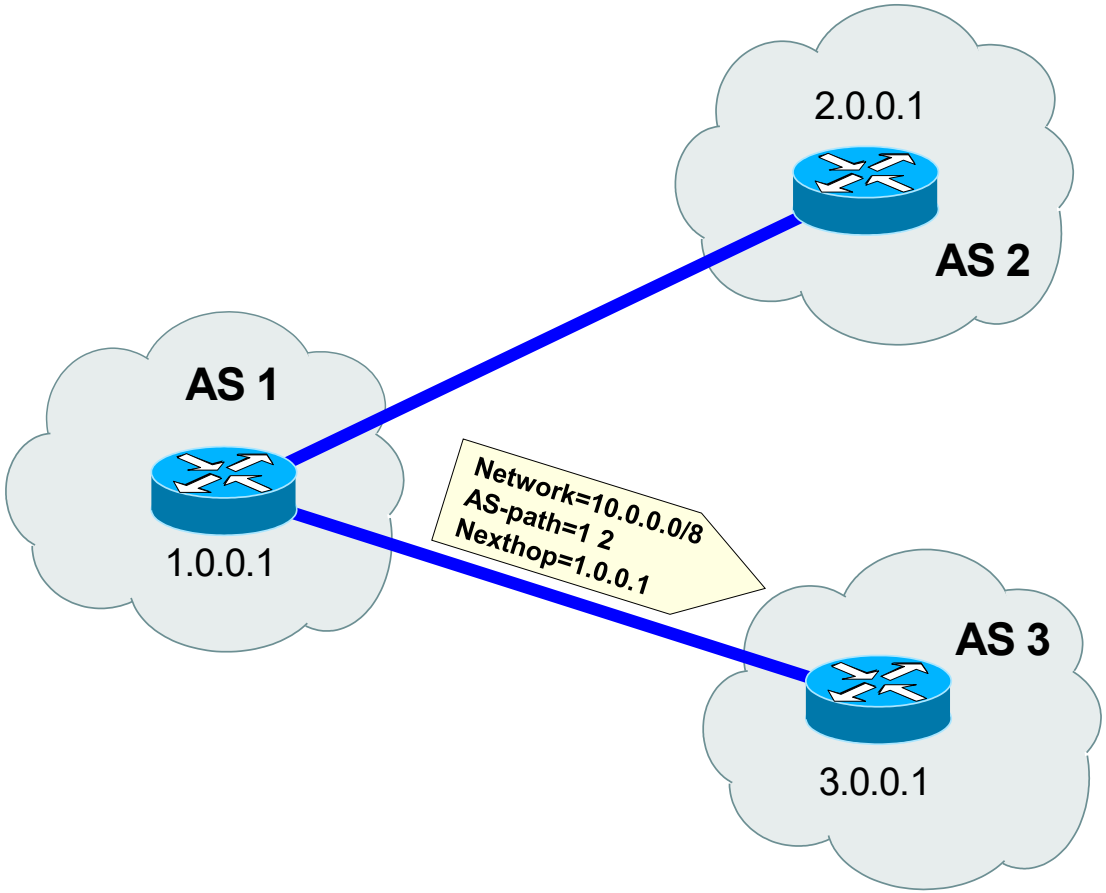
- as 2 router sends an announcement to as 1 specifying 2.0.0.1 as the nexthop



© Computer Networks
Research Group Roma Tre

attributes: nexthop

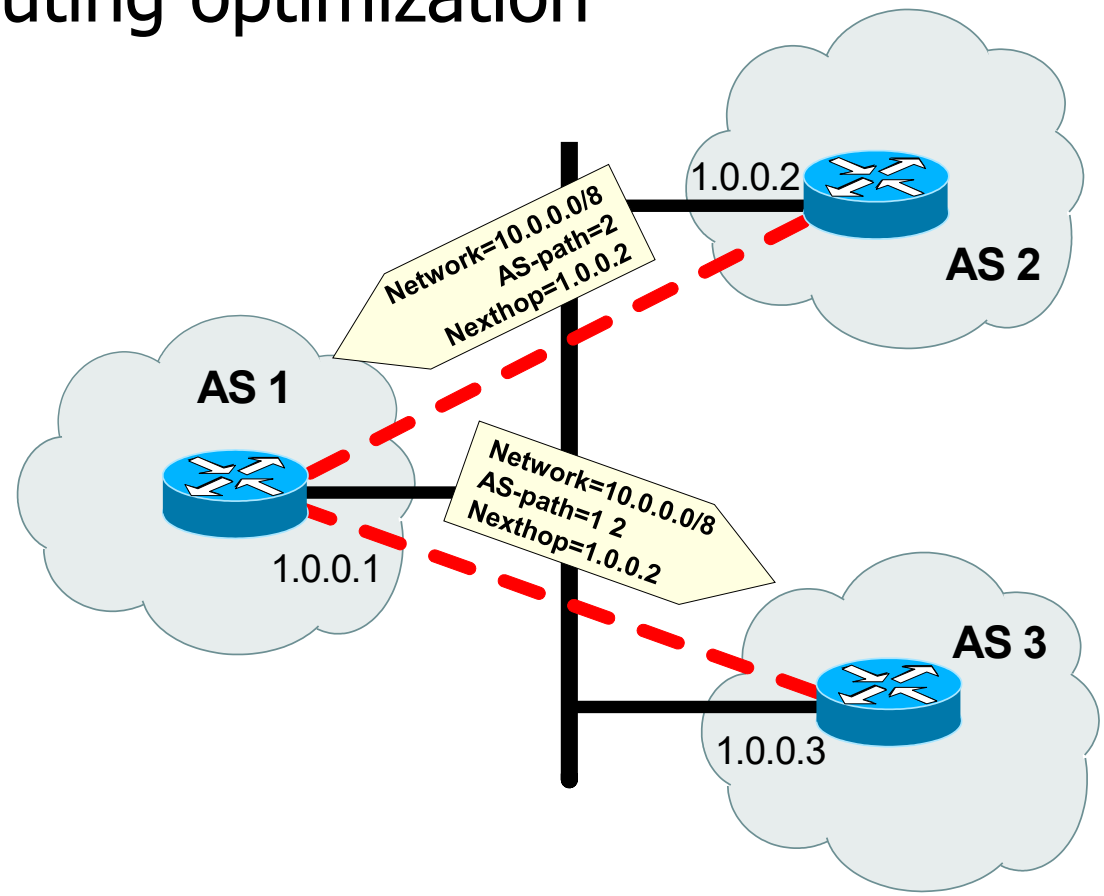
- as 1 router changes the nexthop



© Computer Networks
Research Group Roma Tre

attributes: nexthop

- shared segment: nexthop stays unchanged
  - routing optimization



© Computer Networks
Research Group Roma Tre

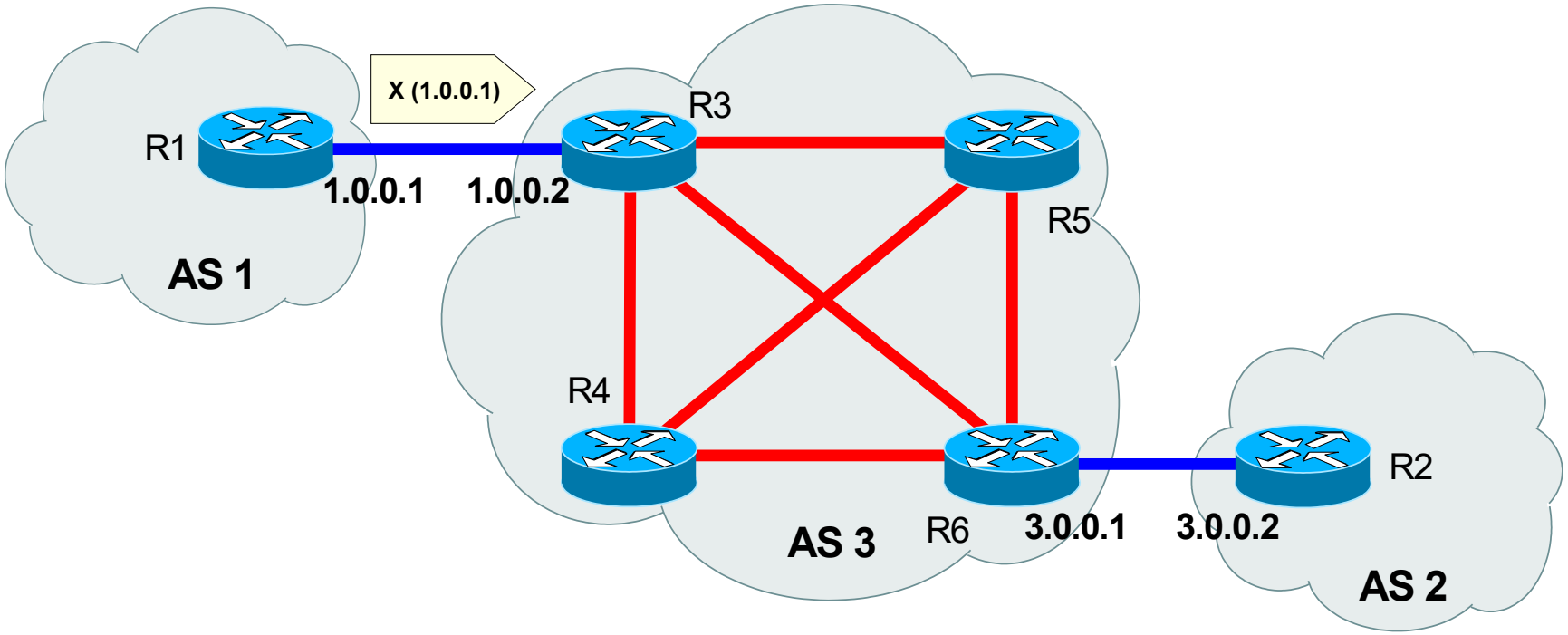
attributes: nexthop

- if a route learned from e-BGP is propagated using i-BGP, then the nexthop remains the same
  - nexthop equal to the address of the remote peer
- internal routers perform a "recursive lookup" for understanding how to reach the nexthop
  - the routers should know, via igp, how to reach the nexthop

© Computer Networks
Research Group Roma Tre

attributes: nexthop

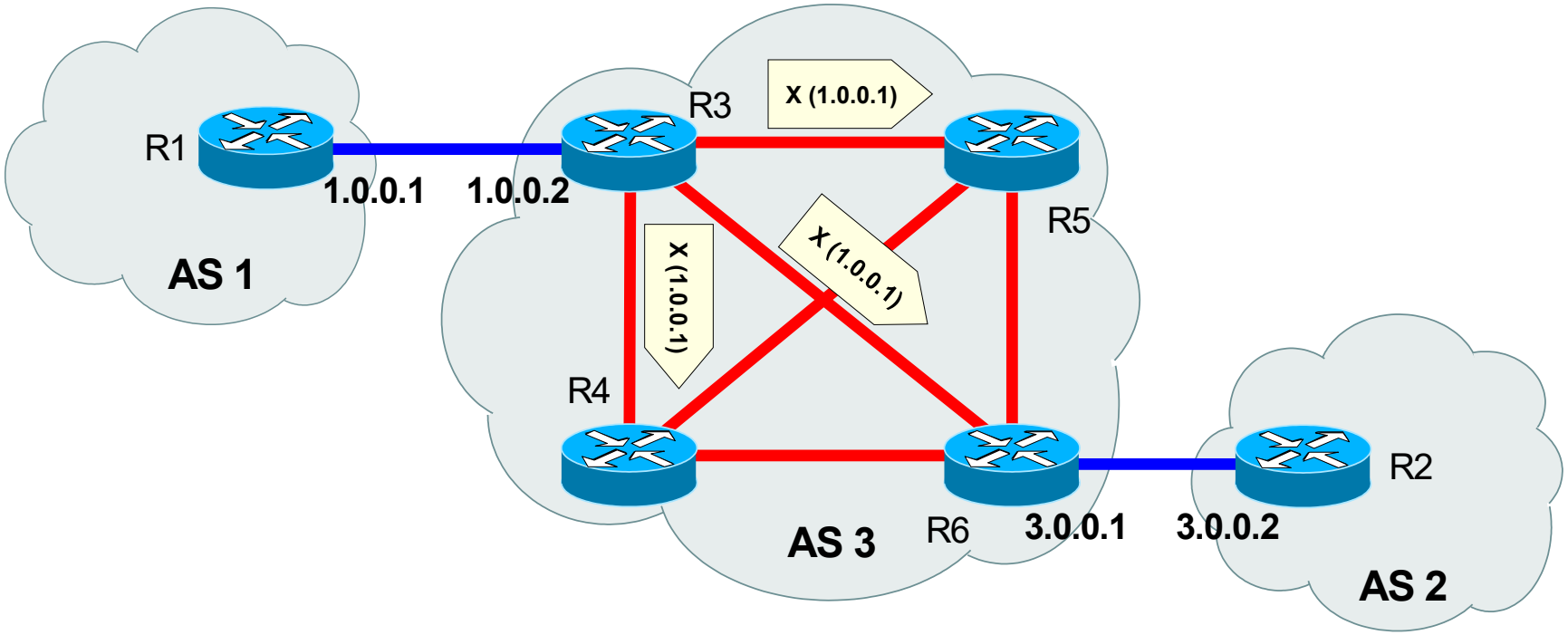
- R1 announces network X with nexthop 1.0.0.1



© Computer Networks
Research Group Roma Tre

attributes: nexthop

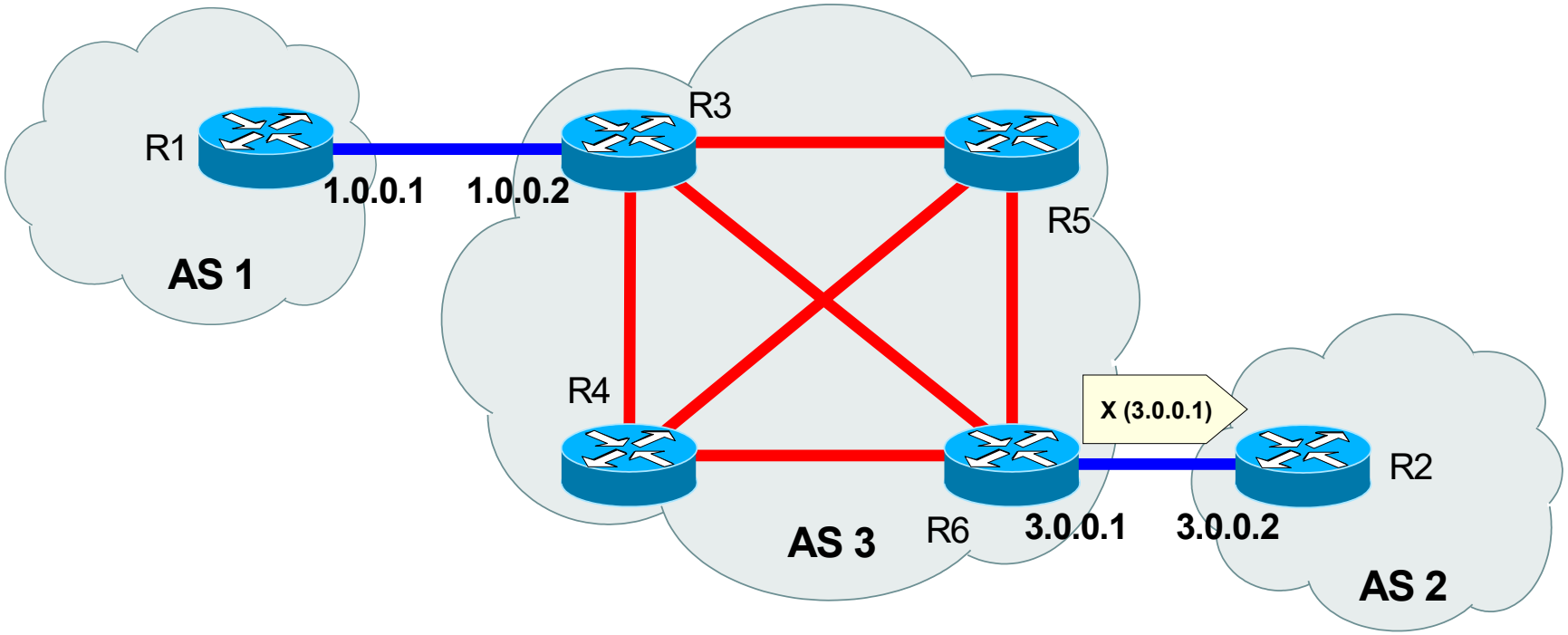
- the announcement is propagated with i-BGP to R4, R5, and R6
  - the nexthop is unchanged



© Computer Networks
Research Group Roma Tre

attributes: nexthop

- the nexthop changes when the announcement goes to a different as



© Computer Networks
Research Group Roma Tre

attributes: origin

- igp
  - declared as internal by the starting AS: “network” command
- egp
  - injected into BGP by EGP: backward compatibility
- incomplete
  - generated redistributing an igp protocol

© Computer Networks
Research Group Roma Tre

path selection

© Computer Networks
Research Group Roma Tre

selection of the “best route” to a prefix

- each router, for each prefix, chooses one of the received announcements as the “best”
- the decision process is fully deterministic
  - no random choice is applied
- only the best routes are (possibly) announced to peers
- selection criteria:
  - more specific and less specific prefixes are considered different prefixes
    - both are injected into the routing table
  - if the next-hop is not reachable (it does not match a line of the routing table of the router) the announcement can not be selected
  - the selection is based both on the values of the attributes and on the constraints imposed by the administrator (e.g. weight)

© Computer Networks
Research Group Roma Tre

selection of the “best route” to a prefix

Weight
Local Preference
Originate
AS Path Length
Origin Type
Multi-Exit Discriminator (MED)
Paths
Router ID

We Love Oranges AS Oranges Mean Pure Refreshment

© Computer Networks
Research Group Roma Tre

© Kevin Wallace

selection of the "best route" to a prefix

Path Selection Parameter	Description
Weight	A locally significant, Cisco-specific parameter that a router can set when receiving updates. A higher Weight is preferred. Commonly used to influence outbound routing decisions.
Local Preference	A parameter communicated throughout a single AS. A higher Local Preference is preferred. Commonly used to influence outbound routing decisions.
Originate	Paths sourced locally are preferred.
AS Path Length	The number of autonomous systems in the AS_PATH path attribute. Lower AS path lengths are preferred. Commonly used to influence inbound routing decisions.
Origin Type	Indicates how the route was injected into BGP: i (`network` command), e (EGP), or ? (redistributed). i is preferred to e, and e is preferred to ?.
Multi-Exit Discriminator (MED)	A parameter set and advertised by routers in one AS to influence the BGP path selection decisions of routers in another AS. A lower MED is preferred.
Paths	Prefer eBGP path over iBGP path.
Router ID	A tie breaker, where the route received from the router with the lowest router ID is preferred.

© Computer Networks
Research Group Roma Tre

© Kevin Wallace

selection of the “best route” to a prefix

(3) Oranges (6) Mean (8) Refreshment (2) Love (1) We (4) AS (5) Oranges

```
R3#show ip bgp
BGP table version is 8, local router ID is 3.3.3.3
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
              r RIB-failure, S Stale, m multipath, b backup-path, f RT-Filter,
              x best-external, a additional-path, c RIB-compressed,
Origin codes: i - IGP, e - EGP, ? - incomplete
RPKI validation codes: V valid, I invalid, N Not found

     Network          Next Hop            Metric LocPrf Weight Path
 *>  5.5.5.0/24       198.51.100.2                        0 65002 65003 i
 *                   198.51.100.6                        0 65004 65003 i
 *>  10.1.1.0/24      0.0.0.0                  0         32768 i
 * i                 10.1.1.2                 0    100      0 i
 *>i 172.16.1.0/24    10.1.1.2                 0    100      0 i
 r>  198.51.100.0/30  198.51.100.2             0             0 65002 i
 r   198.51.100.4/30  198.51.100.2                           0 65002 65003 65004 i
 r>                  198.51.100.6             0             0 65004 i
 *>  203.0.113.0/30   198.51.100.2             0             0 65002 i
 *                   198.51.100.6                           0 65004 65003 i
 *   203.0.113.4/30   198.51.100.2                           0 65002 65003 i
 *>                  198.51.100.6             0             0 65004 i
R3#
```

(7) Pure

© Computer Networks
Research Group Roma Tre

© Kevin Wallace

AS-path filtering (filter-list)

© Computer Networks
Research Group Roma Tre

as-path filtering commands

command syntax

```
neighbor <neighbor-ip> filter-list <acl-name> in
```

command syntax

```
neighbor <neighbor-ip> filter-list <acl-name> out
```

command syntax

```
bgp as-path access-list <acl-name> permit <regexp>
```

command syntax

```
bgp as-path access-list <acl-name> deny <regexp>
```

© Computer Networks
Research Group Roma Tre

as-path filtering commands

- *regexp* may contain the following characters:

`.`	matches any single character
`\`	escapes special characters
`[]`	matches a range of characters
`^`	matches the beginning of a string
`$`	matches the end of a string
`?`	matches zero or one occurrence of a pattern
`*`	matches zero or more occurrences of a pattern
`+`	matches one or more occurrences of a pattern
`()`	groups characters to form a pattern
`|`	matches one of the patterns on either side
`_`	a shortcut for `[,{} ]|^|$`

© Computer Networks
Research Group Roma Tre

as-path filtering example

frr configuration file

```
router bgp 100
network 100.1.1.0/24
neighbor 222.2.2.2 remote-as 200
neighbor 222.2.2.2 filter-list myACL in
!
bgp as-path access-list myACL permit ^200_300
```

- accept from as 200 only the routes received via as 300

© Computer Networks
Research Group Roma Tre

announcement tuning (route-maps)

© Computer Networks
Research Group Roma Tre

attribute setting commands

command syntax

```
neighbor <neighbor-ip> route-map <r-map-name> in
```

command syntax

```
neighbor <neighbor-ip> route-map <r-map-name> out
```

command syntax

```
route-map <r-map-name> permit <seq-number>
  match <announce-property>
  set <attribute-setting>
  ...
```

command syntax

```
route-map <r-map-name> deny <seq-number>
  match <announce-property>
  set <attribute-setting>
  ...
```

© Computer Networks
Research Group Roma Tre

about `route-maps`

- `route-maps` may consist of multiple statements
  - statements are processed in the order established by sequence numbers
  - for each received/sent announcement, only one statement is applied
    - the first one without a `match` condition
    - the first one that matches the announcement attributes (prefix, as-path, etc.)
  - announcements that are not matched by any statement, or that are matched by a `deny` statement are simply filtered out
    - `set` commands in a `route-map deny` are useless
- referencing an undefined `route-map` in a `neighbor` statement results in filtering out everything

© Computer Networks
Research Group Roma Tre

all `match` commands

- match as-path
- match community
- match extcommunity
- match ip address
- match ip next-hop
- match ipv6 address
- match metric
- match origin

© Computer Networks
Research Group Roma Tre

all `set` commands

- set aggregator as
- set as-path prepend
- set atomic-aggregate
- set comm-list
- set community
- set extcommunity
- set ip next-hop
- set ipv6 next-hop
- set local-preference
- set metric
- set origin
- set originator-id
- set weight

© Computer Networks
Research Group Roma Tre

address **match** conditions

- **match ip address** can be used in conjunction with **access-lists** or **prefix-lists**

command syntax

```
match ip address <acl-name>
```

command syntax

```
match ip address prefix-list <prefix-list-name>
```

command syntax

```
access-list <acl-name> permit <network/mask>
```

command syntax

```
access-list <acl-name> deny <network/mask>
```

© Computer Networks
Research Group Roma Tre

about `access-lists`

- an alternative construction to filter prefixes
- the `as-path access-list` variant allows to filter based on as-paths
- `access-lists` are identified by a name or an integer
  - the integer determines the type of filtering applied
    - 1-99: standard access list (filter from specific IPs)
    - 100-199: extended access list (filter by protocol and/or source/destination IP)

© Computer Networks
Research Group Roma Tre

about `access-lists`

- no sequence numbers, still the first matching entry applies; example:
  - `access-list permissiveAcl permit any`
    `access-list permissiveAcl deny any`
    allows everything
  - `access-list restrictiveAcl deny any`
    `access-list restrictiveAcl permit any`
    discards everything
- same for `as-path access-lists`; example:
  - `ip as-path access-list noWay deny .*`
    `ip as-path access-list noWay permit ^100_200`
    discards everything

© Computer Networks
Research Group Roma Tre

`access-list` defaults

- in frr, **`access-lists`** default to **`deny`**
- by default, **`access-lists`** match a prefix as well as all its more specifics; for example:
  - **`access-list myList permit 193.100.0.0/16`**
    also matches **`193.100.5.0/24`**, **`193.100.192.0/25`**, etc.
  - **`access-list permissiveList permit 0.0.0.0/0`**
    matches everything(!)
- this behavior can be changed by using **`exact-match`**
- referencing an undefined **`access-list`** (e.g., in a **`filter-list`** statement) results in filtering out everything

© Computer Networks
Research Group Roma Tre

attribute setting example

frr configuration file

```
router bgp 100
network 100.1.1.0/24
neighbor 222.2.2.2 remote-as 200
neighbor 222.2.2.2 route-map myRouteMap in
!
route-map myRouteMap permit 10
  match ip address myAccessList
  set metric 5
  set local-preference 25
!
route-map myRouteMap permit 20
  set metric 2
!
access-list myAccessList permit 193.204.0.0/16
```

© Computer Networks
Research Group Roma Tre